DAFFODIL INTERNATIONAL UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FYDP (TITLE PHASE) EVALUATION REPORT

REPORTING PERIOD- SUMMER 2025

Project Identification:

I. Project Title	cogneLearn: A Self-Guided Adaptive Learning Platform	
II. Group Members	**ID:** 0242220005101622	**Name:** Md. Abdullah Himel
	ID:	**Name:**
III. Supervisor	**Name:** Md. Jahidul Alam	**Designation:** Lecturer
IV. Co-Supervisor	**Name:** Mehadi Hasan	**Designation:** Lecturer
V. Submission Date:	08-12-2025	
VII. Certificate:	"This is to certify that the final year design project work until Title defense evaluation held on ____________________, titled as stated in *Sec. I*, executed by the students group mentioned in *Sec. II*, have been found satisfactory and every section of this report is reflecting the same."	*(Signature of Supervisor & date)*

CO Description and Corresponding PO for Title Defense Phase

CO	CO Descriptions	PO
CO1	Integrate recently gained and previously acquired knowledge to identify a real-life complex engineering problem for the "**AI-Based Smart Waste Management System**" project.	**PO1**
CO2	Analyze different aspects of the goals in designing an engineering problem solution for the "**AI-Based Smart Waste Management System**" project.	**PO2**
CO3	Explore diverse problem domains through a literature review, delineate the issues, and establish the goals for the FYDP.	**PO4**

1. Introduction:

Online education has grown rapidly, making platforms like YouTube a primary source of knowledge for students. While this access to information is beneficial, it introduces a major challenge: digital distraction. Students often struggle to stay focused because they lack the structured environment of a physical classroom. They face constant interruptions from algorithmic recommendations and notifications. Furthermore, the tools needed for effective study are currently fragmented; a student must constantly switch between a video player for learning and a separate application for time management. This constant switching disrupts concentration and leads to inefficient learning.

To address this issue, this project proposes "cogneLearn: A Self-Guided Adaptive Learning Platform." This is a web-based application designed to help self-directed learners manage their focus. Unlike traditional systems that simply deliver content, cogneLearn combines personal YouTube playlist management with a built-in Pomodoro productivity system. By integrating study materials directly with focus-tracking tools and privacy-safe attention monitoring, the project aims to create a unified environment where students can learn effectively without distractions.

2. Background Study:

The field of online learning tools is vast, but current solutions generally fall into three separate categories. Each category has specific limitations when it comes to self-directed study.

Learning Management Systems (LMS): Platforms like Moodle, Canvas, and Blackboard are the standard for formal education. They are excellent for managing university courses, submitting assignments, and grading. However, these systems are designed for institutions, not for individual learners. They are often rigid and do not easily allow students to import their own external learning materials, such as YouTube playlists or personal tutorials. Most importantly, traditional LMS platforms focus on content delivery but do not provide tools to help students manage their own time or focus while studying.

Productivity and Time-Management Applications: On the other hand, there are many productivity apps available, such as *Focus To-Do* or *Forest* [1]. These applications use techniques like the Pomodoro method to help users stay on task. While effective for time management, they function in isolation. A student must manually set a timer in one application while watching a lecture in another browser tab. This "context switching"—moving back and forth between the learning content and the timer—creates friction and breaks the student's mental flow. Additionally, these tools track time blindly; they have no way of knowing if the user is actually looking at the screen or engaged with the material.

Attention Detection Technologies: Finally, there is the field of computer vision-based attention tracking. Research using libraries like OpenCV has shown that it is possible to track eye movement and head position to estimate focus [2]. However, currently, this technology is almost exclusively used in "online proctoring" software (like *Proctorio*) to prevent cheating during exams [3]. These systems act as surveillance tools, sending sensitive video data to remote servers, which raises significant privacy concerns and causes anxiety for students. There is a lack of tools that use this technology for positive reinforcement—to help the student self-regulate—rather than for policing.

Project Context: cogneLearn is designed to fill the gaps between these three categories. It recognizes that while the technology exists to track focus and manage time, there is no single platform that integrates them for the student's benefit. Existing research also highlights a growing demand for privacy. Users are increasingly wary of cloud-based tracking. Therefore, this project adopts a "privacy-first" approach. By using modern web technologies like "localStorage" and client-side Machine Learning (TensorFlow.js) [4], cogneLearn aims to provide smart analytics and focus tracking without ever sending user behavioral data to a server.

3. Problem Statement:

The core problem this project addresses is the inefficiency and lack of structure in online self-study.

Students who rely on open platforms like YouTube face three critical issues:

1. **Tool Fragmentation:** Learners are forced to use disconnected tools—video players for content and separate mobile apps for time management. This separation disrupts cognitive flow and makes studying less efficient.
2. **Lack of Insight:** Self-learners often lack objective data on their study habits. They may spend hours sitting in front of a screen without realizing they were distracted for most of that time.
3. **Privacy Concerns in Existing Tools:** Current "smart" tools that track attention are designed for surveillance (anti-cheating). They often compromise user privacy by collecting biometric data, making them unsuitable for personal daily use.

Significance & Need for Solution: This problem is significant because passive consumption (often called "tutorial hell") is a major barrier to skill acquisition in the digital age. Without a solution that integrates content with privacy-first productivity metrics, self-directed learners struggle to maintain the discipline required for complex subjects. This leads to wasted time, low retention rates, and the abandonment of learning goals. Therefore, a unified, privacy-conscious platform is urgently needed to transform unstructured video consumption into active, measurable learning.

4. Preliminary Novelty Statement:

The novelty of cogneLearn lies in its integration and its privacy-centric architecture. Uniquely, it binds a custom YouTube playlist manager directly to a synchronized Pomodoro timer, creating a seamless "study-break" cycle within a single window. Furthermore, unlike commercial tools that upload user data to the cloud, this project implements a novel "Client-Side Only" analytics system. By processing productivity stats and machine learning data entirely within the user's browser, it offers intelligent feedback without compromising user privacy.

5. Research Questions:

How can a web-based platform integrate content management with client-side analytics to assist self-directed learners in maintaining focus without compromising their data privacy?

6. Objectives:

The key objectives of this project are:

I. To develop a full-stack web application using Java Spring Boot and React.js that allows users to manage distraction-free YouTube study playlists.
II. To implement a Pomodoro Productivity System that tracks study sessions and visualizes productivity trends using a privacy-secure, local storage mechanism.
III. To integrate a "Mujahid Mode" (Focus Mode) that guides users to block external distractions effectively.
IV. To demonstrate technical capability by integrating a browser-based Machine Learning model to detect real-time attention levels via webcam, providing immediate feedback to the user.

7. Expected Outcome:

The project will result in a fully functional "Adaptive Self-Study Platform" deployed as a web application.

- **Practical Result:** A tool where students can log in, access their study playlists, and engage in tracked study sessions with a dashboard showing their productivity history.

- **Technical Contribution:** A demonstration of a "privacy-first" EdTech architecture, proving that meaningful behavioral analytics and attention tracking can be implemented entirely on the client-side without collecting sensitive user data on a central server.

8. Scope of Work:

Boundaries: The scope of this project is limited to the development of a web-based "Adaptive Self-Study Platform."

- **User Management:** The system will handle basic user registration and authentication for individual learners. Institutional (admin/teacher) roles and multi-user classroom features are outside the current scope.
- **Content:** The platform will support importing and playing YouTube playlists only. Hosting original video files or integrating other video providers (like Vimeo or Dailymotion) is not included.
- **Analytics:** Productivity data will be stored strictly on the client-side (browser "localStorage") for privacy reasons. There will be no centralized server-side data warehousing or cross-user data comparison.
- **Attention Tracking:** The machine learning component is an experimental feature designed for real-time feedback. It will use a pre-trained model for inference; training a new custom model from scratch is beyond the scope of this project.

Limitations:

- The attention tracking feature depends on webcam quality and lighting conditions; poor environments may affect accuracy.
- The "Mujahid Mode" (distraction blocking) relies on user compliance to install suggested extensions, as a web application cannot force-install browser add-ons due to security sandboxing.
- The application requires an active internet connection to stream YouTube content.

9. References:

[1] F. Cirillo, *The Pomodoro Technique: The Acclaimed Time-Management System That Has Transformed How We Work*. New York, NY, USA: Currency, 2018.

[2] A. Das, U. Gawde, K. Suratwala, and D. Kalbande, "Significance of Facial Landmarks for Student Engagement Detection," in *2020 3rd International Conference on Intelligent Sustainable Systems (ICISS)*, Thoothukudi, India, 2020, pp. 367-372.

[3] S. R. G. and S. M., "Real-Time Driver Drowsiness Detection System using Eye Aspect Ratio," *International Journal of Engineering Research & Technology (IJERT)*, vol. 9, no. 06, pp. 45-49, 2020.

[4] V. Mandic, "Human: AI-powered 3D Face Detection & Rotation Tracking," GitHub repository, 2023. [Online]. Available: https://github.com/vladmandic/human. [Accessed: Dec. 08, 2025].

[5] J. Long, "Building a RESTful Web Service," *Spring Guides*, 2024. [Online]. Available: https://spring.io/guides/gs/rest-service/. [Accessed: Dec. 08, 2025].

Supervisor's Feedback: Provide comments and suggestions for improvement
1. ____________________________ **2.** ____________________________ **3.** ____________________________

Examiner's Signature: __________________________

Date: __________________________